

National Exam Of Higher National Diploma 2022 Session

Spécialty/Option: Software Engineering (SWE)
Paper : System Analysis And Design
Duration : 4 hours

Credits :7

SECTION A: INFORMATION SYSTEM AND DATA BASE (50MARKS)

I. Object Modeling 25 Marks)

- 1- Differentiate between a class and a method. (5marks)
- 2- What is the difference between public, private and protected access modifiers? (5marks)
- 3- List some characteristics of UML? (5marks)
- 4- Class diagrams are the most common diagrams used in UML. Explain four purpose of the class diagram. (5marks)
- 5- Give an example of a class, composed of a name, attributes and methods. (5marks)

II. Database Administration with MySQL (25 Marks)

- 1- Explain the following terms briefly: one-to-many relationship, many-to-many relationship, participation constraint, overlap constraint. (4marks)
- 2- List and explain four advantages of using a DBMS (3marks)
- 3- Does the relational model, as seen by an SQL query writer, provide physical and logical data independence? Explain. (3marks)
- 4- Why does a DBMS interleave the actions of different transactions instead of executing transactions one after the other? (5marks)

FIELDS (ATTRIBUTES, COLUMNS)

Field names →

<i>sid</i>	<i>name</i>	<i>login</i>	<i>age</i>	<i>gpa</i>
50000	Dave	dave@cs	19	3.3
53666	Jones	jones@cs	18	3.4
53688	Smith	smith@ee	18	3.2
53650	Smith	smith@math	19	3.8
53831	Madayan	madayan@music	11	1.8
53832	Guldu	guldu@music	12	2.0

← TUPLES (RECORDS, ROWS)

- 5- Consider the instance of the Students relation shown in Figure above.
- Give an example of an attribute (or set of attributes) that you can deduce is not a candidate key, based on this instance being legal. (5marks)
 - Is there any example of an attribute (or set of attributes) that you can deduce is a candidate key, based on this instance being legal? (5marks)

SECTION B: WEB AND MOBILE PROGRAMMING (50MARKS)

I. Web Programming (25 Marks)

- What are the advantages / disadvantages of various style methods? (5marks)
- List some necessities of using HTML forms. (5marks)
- How do I create frames? What is a frameset? (5marks)
- What are the form elements that support media content in HTML5?(5marks)
- How can page loading time be reduced? (5marks)

II. Mobile Programming (25 Marks)

- What Is the Google Android SDK? (5marks)
- Describe the Android Framework. (5marks)
- What is the importance of having an emulator within the Android environment? (5marks)
- Enumerate the three key loops when monitoring an activity. (5marks)
- Explain the proper way of setting up an Android-powered device for application development. (5marks)